

Overview

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 |
Generated: | Engine: DiscreteSim | Sections: 8

Overview

A discrete-event simulation study: the engine, the calibration and the experiments, with every number carrying its confidence interval.

MODELS
5

THROUGHPUT
0.133 items_per_hour

CYCLE TIME
MEAN
552.5 minutes

UTILISATION
0.863

COMPLETED
477.5

VALIDATION
validated

REEVALUATION
keep

Design of the study

Seed 20260101
Days 45
Replications 8
Horizon 4000
Warmup 400
Models [:mmc, :transfer_line, :machine_shop, :inventory, :call_center]
Objective cycle_time_mean
Online policy offline_first

The models

Catalogue

Model	Title	Entity	Resource	Answers
mmc	ServicePool	customer	server	ServicePoolWithIdenticalServers
transfer_line	TransferLine	part	station	SerialLineWithBuffersAndFailures
machine_shop	JobShop	job	machine	JobShopWithRoutingsReworkScrapAndBreakdowns
inventory	InventoryPosition	order	stock	ReorderPointPolicyWithLeadTimeAndBackorders
call_center	ContactCenter	call	agent	AgentPoolWithPatienceAndPriority

Resources

Resource	Kind	Servers	Discipline	Utilisation	Availability	Queue length	Wait	Service	Served	Breakdowns
mill	machine	1	priority	0.904	1	5.674	65.8	9.771	332	0
drill	machine	1	priority	1	1	33.6	277.8	8.2	440	0
grinder	machine	1	priority	1	1	35.1	315.7	9.682	371	0

Resource	Kind	Servers	Discipline	Utilisation	Availability	Queue length	Wait	Service	Served	Breakdowns
lathe	machine	1	priority	0.742	1	0.793	9.547	8.957	299	0
press	machine	1	priority	0.589	1	0.479	6.022	7.399	286	0

Machine shop over 8 replications

Metric	Mean	Half width	Unit	Rel hw pct
wait_mean	552.5	12.7	minutes	2.294
wait_p95	1034	103.3	minutes	9.987
cycle_time_mean	552.5	12.7	minutes	2.294
service_mean	—	—	minutes	—
completed	477.5	9.543	count	1.998
throughput	0.133	0.00265	items_per_hour	1.998
wip_mean	4.155	0.061	count	1.476
tardiness_mean	449	11.5	minutes	2.564
scrapped	22.9	4.057	count	17.7
reworked	71.2	5.031	count	7.061
setup	2	0	minutes	0
utilisation	0.863	0.011	ratio	1.285
availability	1	0	ratio	0
max_queue_length	44.1	6.49	count	14.7
queue_length_mean	75.5	0.054	count	0.071
span	3599	0.388	count	0.011
events	7762	105.8	count	1.363